Global Superstore Sales & Profit Analysis: Project Report

Oct 13, 2026 · @Ayush Yadav

Project Overview

Global Superstore earned $12.64M in sales and $1.47M in profit between 2011 and 2014, an 11.6% profit ratio across 25,035 orders. This project cleans the order data in Python and presents it in two interactive Tableau dashboards to show where the business makes money and where it loses it.

The analysis covers four years of orders from 147 countries and 4,873 customers. It was built as a portfolio project to practise an end-to-end analyst workflow: cleaning, feature engineering, visualisation and written insights.

Business Objectives

The analysis answers six questions about what drives and what hurts profit:

- Which categories and sub-categories earn the most profit, and which lose money?
- Which countries and markets are the most and least profitable?
- Who are the most and least valuable customers?
- How does discounting affect profit?
- How have sales and profit changed from 2011 to 2014?
- How long does each shipping mode take?

Dataset

The raw file `superstore.csv` has 51,290 rows and 27 columns, one row per order line, covering orders placed from 1 January 2011 to 31 December 2014. After cleaning and feature engineering, `superstore_clean.csv` has 51,290 rows and 35 columns.

Group	Key columns
Orders	order_id, order_date, ship_date, ship_mode, order_priority
Customers	customer_id, customer_name, segment
Location	country, state, city, region, market
Products	category, sub_category, product_name
Measures	sales, quantity, discount, profit, shipping_cost

Tools and Technologies

Tool	Purpose
Python (pandas)	Cleaning, date handling, and new analysis columns
Jupyter Notebook in VS Code	Writing and running the cleaning code
Tableau Public	Two interactive dashboards
GitHub	Hosting the code, data, and dashboards

Data Cleaning and Preparation

The notebook `superstore.ipynb` cleans the raw file in nine steps and found no missing values, duplicate rows, or invalid dates, so no rows were removed.

1. Loaded the raw data (51,290 rows, 27 columns).
2. Renamed the Chinese record-count column to English and standardised all column names to lowercase with underscores.
3. Converted `order_date` and `ship_date` to dates.
4. Checked for nulls, duplicate rows, duplicate row IDs, and orders shipped before they were placed. All checks returned zero.
5. Dropped duplicates and the constant `record_count` column.
6. Trimmed extra spaces from text columns.
7. Created new columns: order year, month, quarter, year-month, day of week, shipping days, profit margin %, a loss flag, and discount bands (No discount, 1-10%, 11-20%, 21-30%, 31-50%, 50%+).
8. Dropped the original `year` column, which duplicated `order_year`.
9. Saved the result as `superstore_clean.csv` for Tableau.

The discount bands and shipping days columns power two of the dashboard charts.

Dashboards

Two linked Tableau dashboards cover the full 2011 to 2014 period, with filters for category, market, order year, and segment.

Overview shows four headline numbers (total sales, total profit, profit ratio, total orders), monthly sales and profit trends, profit by discount band, a world map of profit and loss by country, and profit margin for every sub-category.

Deep Dive (Countries, Customers and Shipping) shows the top 10 countries by profit, the 10 most loss-making countries, the top 10 customers and the 10 customers who lost the most, average shipping days by ship mode, and a scatter of average discount against profit by sub-category.

Key Findings

Discounts above 20% drive the most lost profit, and a few countries and one sub-category account for most of the rest.

1. Sales and profit grew every year

Sales rose 90%, and profit rose 103% from 2011 to 2014, so the profit ratio improved slightly as the business grew.

Year	Sales ($M)	Profit ($K)
2011	2.26	249
2012	2.68	307
2013	3.41	407
2014	4.30	504

2. Discounts above 20% make the business lose money

Orders with no discount earned $1.77M in profit, while orders discounted 31% or more lost about $0.79M between them. The 21-30% band is already slightly negative, and the scatter on the Deep Dive dashboard shows sub-categories with higher average discounts earn less profit.

Discount band	Sales ($M)	Profit ($K)
No discount	6.99	1,771
1-10%	1.96	338
11-20%	1.76	173
21-30%	0.38	-21
31-50%	1.18	-381
50%+	0.37	-413

3. Tables is the only losing sub-category

Tables sold $757K but lost $64K, a margin of -8.5%. Every other sub-category is profitable. Copiers ($259K) and Phones ($217K) earn the most profit. Tables pull Furniture down to a 6.9% margin, the weakest of the three categories.

Category	Sales ($M)	Profit ($K)	Profit ratio
Technology	4.74	664	14.0%
Office Supplies	3.79	518	13.7%

Category	Sales ($M)	Profit ($K)	Profit ratio
Furniture	4.11	285	6.9%

4. Turkey and Nigeria lose the most

Turkey (-$98K), Nigeria (-$81K), and the Netherlands (-$41K) are the biggest loss-making countries. The United States earns the most profit at $286K, followed by China ($151K) and India ($129K). By market, EMEA has the lowest profit ratio at 5.4%.

5. Customers and shipping

The most profitable customer, Tamara Chand, earned about $8.7K, while the biggest loser, Cindy Stewart, lost about $6.2K. Standard Class takes about 5 days to ship, Second Class about 3.2, First Class about 2.2, and Same Day under a day. Consumer, Corporate, and Home Office segments all have similar profit ratios of 11.5% to 12.0%.

Recommendations

1. **Cap discounts at 20%.** Orders above that level lost money in every band, so approval should be required for anything higher.
2. **Review Tables pricing and discounts.** It is the only sub-category with a loss, and its average discount is the highest on the scatter chart (about 29%).
3. **Investigate Turkey and Nigeria.** Together they lost about $179K. Check whether discounts, shipping costs, or product mix explain it before deciding to reprice or exit.
4. **Look into EMEA.** Its 5.4% profit ratio is well below the 10-14% of other large markets.
5. **Protect the strong performers.** Copiers, Phones, and Accessories combine high profit with healthy margins of 13% to 17%.

Conclusion and Next Steps

Global Superstore is growing and profitable at 11.6%, but heavy discounting, one product line, and a handful of countries give back a large share of that profit. Fixing the discount policy is the clearest and most controllable lever.

Possible next steps:

- Analyse shipping cost against profit by country, to explain the Turkey and Nigeria losses.
- Add a SQL layer, loading the cleaned data into a database and writing the business queries there.
- Build a customer segmentation, such as RFM, to go beyond the top and bottom 10 lists.

Project files and dashboards are on GitHub in the global-superstore-sales-profit-analysis repository.